

Institutional Report - Viçosa

General Analysis

Viçosa combines an active university base, well-established connectivity infrastructure, and functioning digital public services, and this combination is reflected in more than one area of municipal management. The presence of tecnoPARQ and the Federal University of Viçosa (UFV), the concentration of technology companies, and internet coverage above the expected level create tangible conditions that go beyond the economic sector: they support vocational training initiatives, sustain the provision of online services by the municipal government, and have paved the way for formal cooperation agreements in urban innovation, such as the Cidades do Futuro program and the establishment of the Viçosa Smart Laboratory. In the social sphere, internet coverage in schools and the use of electronic medical records in Family Health units show that technology has already reached essential sectors. In infrastructure, high rates of sewage collection and urban waste collection indicate that the municipality organizes basic services with relative consistency. A reasonable level of budget transparency and the availability of public data complete a picture in which there is, in fact, a solid base on which to advance.

The main pattern across the four dimensions is the gap between what already exists and the municipality's capacity to manage, integrate, and sustain what has been built. The municipality offers digital services, but does not measure whether they effectively reach citizens. It collects sewage, but still treats only a minimal fraction. It has technology companies and an innovation park, but the municipal administration does not have a digital transformation plan or an IT unit sized to coordinate this process. It uses electronic medical records in health services, but the systems do not communicate with one another. It has signed cooperation agreements on smart cities, but without defined deadlines and responsibilities. In all these situations, the problem is not the absence of initiatives, but the lack of integration among them and of an institutional structure capable of connecting and guiding them. This fragmentation also appears in environmental management, where the absence of real-time monitoring leaves the municipal government reacting to problems instead of anticipating them, and in public safety, where the existing instruments have not been sufficient to reverse a concerning trend. Before expanding what already exists, the municipality's next stage of development requires building the missing management layer: integrated planning, technology governance, results monitoring, and effective channels for public participation. Without these foundations, the progress already achieved remains vulnerable to discontinuity.

Economic Dimension

Viçosa has an economic base concentrated in the service sector, with public administration also accounting for a significant share of local activity. The municipality has a substantial number of information and communication technology (ICT) companies, a sector that includes hardware, software, and digital services, as well as formal jobs in this area, indicating a productive structure with a certain degree of technological density for a medium-sized city. The presence of the Viçosa Technology Park (tecnoPARQ), linked to the Federal University of Viçosa (UFV), demonstrates organized innovation initiatives in the territory operating at a relevant scale: the park is actively expanding, generates a significant economic impact in the city, and has already formalized international cooperation agreements. Mobile internet coverage is above the expected average for municipalities of this size, and fiber-optic infrastructure already reaches the municipality, creating a connectivity base that can serve both businesses and the public administration. Sanitation and waste collection indicators are also high, contributing to an urban environment with adequate basic conditions for economic activity.

Despite the presence of ICT companies and an active innovation ecosystem, access to credit and microcredit for entrepreneurs and small businesses does not appear in the indicators analyzed. The municipality has advanced in simplifying the business environment through the implementation of the integrated network for business registration and legalization and the operation of the Sala Mineira do Empreendedor, but there is no active initiative providing accessible financial credit for people who want to start or maintain a

business. This gap particularly affects individual microentrepreneurs and local small businesses. Urban productive inclusion and employment and income-generation indicators are at intermediate levels, suggesting that part of the population still faces difficulties entering the formal labor market. Vocational training and job placement are also at moderate levels, although tecnoPARQ, Senac, and the municipal government already maintain training programs in operation, which indicates that there is a foundation for expanding the reach of these initiatives. The municipality's payment capacity is not available in the data, making it difficult to assess fiscal room for new investments, but the other indicators suggest there is scope for more active local economic development policies.

Another point requiring attention is the gap between the potential of the local innovation ecosystem and the effective use of digital tools in municipal management and services. The municipality does not have an integrated smart-city platform, meaning a system that centralizes and connects different urban services and data, and the digital services offered through the municipal website are still limited. There is, however, institutional coordination underway: the municipal government signed the Cidades do Futuro technical cooperation agreement with the Government of Minas Gerais, associated with the inauguration of the Viçosa Smart Laboratory at tecnoPARQ. This provides a concrete starting point for the integrated digitalization of urban management. It also indicates that the technology companies and institutions present in the territory could be used more effectively to solve practical public-administration challenges. This coordination already has an institutional channel, but it still lacks a structured plan with defined deadlines, responsibilities, and priorities so that it can translate into concrete results for the population.

Improvement recommendations:

- Create a municipal microcredit access program for individual microentrepreneurs and small businesses, with particular focus on the technology and service sectors. The municipality does not record any active initiative offering accessible financial credit in this area, and this gap constrains the ability of local entrepreneurs to maintain or expand their businesses. The Sala Mineira do Empreendedor, which is already operating, can serve as the natural channel for coordinating credit lines with state and federal development agencies, without the need to create a new facility from scratch.
- Strengthen and expand the existing partnership among the municipal government, tecnoPARQ, and Senac for vocational training, increasing the reach of programs focused on technology and digital services. Vocational training and job-placement indicators are at an intermediate level, suggesting that part of the working-age population still does not have access to training or re-employment opportunities. Scaling up initiatives already in operation and connecting them more directly to the actual needs of local ICT companies can increase employability and reduce the gap between labor supply and demand.
- Use the Cidades do Futuro technical cooperation agreement as a starting point to map the municipal administration's digital needs and make them accessible to local technology companies. With a significant number of ICT companies in the municipality and tecnoPARQ expanding, there are concrete conditions for developing solutions tailored to urban-management problems without relying exclusively on external suppliers. This mapping can be carried out in a simple way, with participation from different municipal departments, and can lead to more precise public calls or procurement processes.
- Develop a municipal digital transformation plan that guides the digitalization of public services and the use of data in city management, with defined deadlines and responsibilities, using the Cidades do Futuro agreement as an institutional starting point. The absence of an integrated urban-services platform and the still limited level of digital services on the municipal website indicate that digitalization is taking place in a fragmented way. A structured plan aligned with the municipality's Master Plan would make it possible to prioritize investments, avoid duplicated efforts, and make services more accessible to the population.

Sociocultural Dimension

Viçosa has an educational structure with very broad internet coverage in municipal public schools, and the number of computers available to students provides tangible support for digital learning. The municipality

has a reasonable number of cultural and sports facilities for a medium-sized city, as well as records of protection for tangible and intangible cultural heritage. The illiteracy rate is below the more critical levels observed in other regions, and performance in IDEB, the Basic Education Development Index, for the final years of elementary education is at a moderate level. The municipality's high Human Development Index reinforces the existence of relatively favorable social conditions compared with the national average.

Despite good internet coverage in schools, the absence of educational initiatives aimed at specific communities indicates that more vulnerable groups are not being reached by digital-inclusion or technological-literacy programs. The age-grade distortion rate, which measures the gap between a student's age and the school year they should be attending, remains significant, indicating that some students face difficulties that go beyond the infrastructure available. In health care, the indicators point to an important gap: although Family Health Strategy units already use electronic medical records, this system is not yet integrated with the other units in the municipal public network, and no digital health services are offered to patients, such as online appointment scheduling or access to test results. This means that health management still operates separately from the possibilities already offered by technology, despite the significant number of physicians and hospital beds in the network. The high Gini index, which measures income inequality, reinforces the fact that existing progress does not reach all residents equally.

Public safety and risk management also require attention. The municipality has some monitoring solutions and a set of public-safety policies, but the homicide rate remains concerning for a medium-sized municipality with a relatively high Human Development Index, and the recent historical series shows a consistently worsening trajectory, with the number of cases more than doubling over the last five years. There is a contrast between the existence of natural-disaster management instruments and the low availability of online cultural and public-participation services, indicating that the digitalization of citizen services is progressing unevenly across policy areas. Existing digital participation channels, such as the electronic ombudsman service and live broadcasts of legislative sessions, cover basic functions, but more robust instruments are lacking, such as public-consultation platforms and structured proposal-submission mechanisms, especially within the municipal Executive Branch. This limitation reduces community involvement in local decisions and the municipality's ability to identify territorial demands quickly.

Improvement recommendations:

- Create a municipal digital-inclusion program specifically aimed at vulnerable groups, such as older adults, people with low levels of schooling, and residents of peripheral areas, with in-person activities in easily accessible locations such as schools, community centers, and existing cultural facilities. The complete absence of recorded actions for specific communities and the high income-inequality index justify this priority.
- Expand and integrate the electronic medical record system already used in Family Health Strategy units across the other units of the municipal public network, and develop at least one digital health service for patients, such as online appointment scheduling or access to test results. Integration between units is still in the planning stage, and the absence of any patient-facing digital health service represents an operational gap that undermines quality of care and continuity of treatment, especially given the significant number of physicians and hospital beds already available in the network.
- Expand digital forms of public participation in municipal decision-making through accessible platforms for public consultation, submission of suggestions, and monitoring of local policies. Existing digital channels, including the electronic ombudsman service and broadcasts of legislative sessions, cover basic functions, but the municipality still relies heavily on in-person formats and lacks more robust instruments, especially within the municipal Executive Branch. This limits citizen engagement among people who cannot attend meetings and public hearings.
- Develop an ongoing technology-training program for educators in the municipal public-school network, with the aim of enabling them to work with digital literacy in the classroom in a practical way that is integrated into the curriculum. With only one technology-education center and two recorded technology-training courses, there is a clear gap between the infrastructure available in schools, including computers and internet access, and the pedagogical capacity to use it consistently.

Environmental Dimension

In sanitation, Viçosa records a sewage-collection rate covering most of the urban population, a significant result for a medium-sized municipality. Average water consumption per inhabitant is at a moderate level, within a range considered reasonable for cities of similar size. Protection and management of green areas also receive a partial score, indicating that some institutional structure is dedicated to environmental issues. In this respect, it is worth noting that the Municipal Department of Environment, Biodiversity and Water Resources (SEMABIO) also assumed responsibility for the management of parks and gardens following legislative approval in 2026, making the institutional structure more cohesive than the indicators alone might suggest.

The most serious challenges are in the sanitation chain and the absence of monitoring tools. The volume of sewage treated is virtually zero in the period measured by the indicators, which means that most collected sewage was discharged without adequate treatment, a serious problem for public health and the region's rivers. There is a concrete initiative underway in this area: the Barrinha Wastewater Treatment Plant, a project of the Viçosa Autonomous Water and Sewage Service (SAAE Viçosa), was at an advanced stage of construction with expected entry into operation by the end of 2024, but there is no official confirmation that it is already operating at full capacity. Until this facility is completed and fully operational, the problem persists. Water-distribution losses are also high, indicating that more than one third of the water produced does not reach the end consumer, whether due to leaks, irregular connections, or network failures. Selective collection accounts for only a very small share of total solid waste collected, pointing to a system that is still underdeveloped. However, in May 2026 the municipality launched the Viçosa Recicla program, coordinated by SEMABIO, with the aim of expanding selective collection and integrating the waste-picker associations ACAMARE (Association of Workers of the Viçosa Sorting and Recycling Plant) and ACAT (Association of Recyclable-Material Collectors of Viçosa). These three problems, untreated sewage, water losses, and low recycling rates, have direct impacts on the environment and on the cost of public services.

Another aspect requiring attention is the complete absence of digital solutions applied to environmental and infrastructure management. There are no tools to monitor water consumption, optimize waste collection, track air quality, control gas emissions, or remotely manage public lighting. This technological gap is not merely a modernization issue: without real-time data, the municipal government is limited to reactive management, responding to problems after they occur instead of preventing them. Gradual adoption of monitoring solutions, even on a limited scale, could improve the efficiency of existing services and provide a stronger evidence base for planning decisions.

Improvement recommendations:

- Closely monitor the start of operations at the Barrinha Wastewater Treatment Plant and ensure that SAAE Viçosa provides regular information on treatment performance. Once the plant is fully operational, it will be important to monitor its efficiency and plan the gradual expansion of capacity so that all collected sewage is properly treated, in compliance with sanitary standards and with reduced contamination of the region's rivers. Coordination with the Intermunicipal Regulatory Agency for Sanitation Services of the Zona da Mata of Minas Gerais and Adjacent Areas (ARIS-MG) can support this monitoring.
- Develop a structured program to reduce losses in the water-distribution system, beginning with a detailed network diagnosis to identify sections with the highest leakage rates and irregular connections. Reducing losses improves system efficiency and lowers operating costs, freeing resources that can be reinvested in the network itself. Simple sectorized metering tools can already identify where losses are greatest, even without major technological investments.
- Monitor and expand the recently launched Viçosa Recicla program coordinated by SEMABIO, rather than proposing the creation of a new program. The immediate priority should be to ensure that selective-collection routes reach a significantly larger share of the population, strengthen working conditions for the ACAMARE and ACAT associations, and establish measurable performance targets.

Environmental-education campaigns in schools and in neighborhoods with lower participation can also help increase public engagement and the volume of recyclable material diverted from landfills.

- Begin adopting digital monitoring in at least one priority area of environmental management, such as water consumption or public lighting, with the objective of building a data base to guide future decisions. Remote-management solutions for public lighting, for example, can identify failures and adjust network operation remotely, reducing costs and energy waste. This first step can serve as a model for gradually expanding the use of data in the management of other environmental services.

Institutional Capacities Dimension

Viçosa already has a functional base for providing public services online, with a significant number of services available on the internet and many of them allowing citizens to submit requests digitally. Financial and budget transparency also performs reasonably well, and the availability of data to the public is relatively broad. Hardware and software storage infrastructure is at an acceptable level for a medium-sized municipality, supporting existing digital operations. Collaborative governance has identified responsible actors, and the areas of education, health, public works, the Municipal Department of Planning, Mobility and Territorial Management, which covers urban planning, mobility and transport, and the Municipal Department of Financial Management are already recognized as priority areas for the use of technology.

The most critical gaps are precisely in the foundations needed to support consistent progress in digital management. The municipality has no strategic planning focused on digital transformation and no formalized information technology (IT) governance practices, which means that existing initiatives tend to occur in isolation, without clear direction or shared criteria across departments. The restructuring of the municipal organizational chart carried out in January 2025 modernized departments and reorganized responsibilities, but did not include the creation of a formal unit dedicated to information and communication technology (ICT), confirming and updating this gap. The IT team is very small relative to the demands of a municipality with this breadth of digital services. Integrated data management and monitoring of both public safety and policy transparency are at very low levels, weakening the municipality's ability to make decisions based on reliable and up-to-date information. In addition, public perception of municipal services is not being measured, preventing the administration from knowing whether the digital services offered are effectively meeting citizens' needs.

This set of weaknesses reveals a pattern common to municipalities that have advanced in the digitalization of services without building, in parallel, the institutional structure needed to sustain the process. Having online services in operation is important, but without planning, governance, and monitoring, these services remain vulnerable to discontinuity, inconsistency, and low quality. The lack of integration among data produced by different departments also limits the municipality's ability to understand problems comprehensively and act in a coordinated way. It is worth noting that adoption of the national Electronic Services Invoice standard (NFS-e), currently being implemented, provides a concrete starting point for data integration in the tax module, although it is limited in scope and stems from a federal legal requirement rather than an autonomous digital-transformation initiative. For a medium-sized municipality with responsibilities in the five identified priority areas, the misalignment between the existing digital-service offering and the management structure supporting it is the main point of concern in this dimension.

Improvement recommendations:

- Develop a municipal digital transformation plan establishing priorities, targets, and responsibilities for areas that already use technology, such as education, health, public works, urban planning, and transport. The plan should be formally approved by the administration and reviewed periodically, serving as a reference for technology-investment decisions.
- Create a formal ICT organizational structure within the municipal administration, with defined responsibilities and a team sized according to the municipality's actual demand. The organizational restructuring carried out in 2025 did not address this gap, making the measure even more urgent. This

structure should have the authority to coordinate digital initiatives across different municipal departments and establish minimum information-technology governance practices.

- Implement an integrated data-management system connecting information produced by the main municipal departments, allowing the administration to consult and cross-reference data from different policy areas in a single environment. Adoption of the national NFS-e standard, already underway in the tax module, can serve as a starting point for broader integration. This can reduce duplicated work, improve the quality of information used in decision-making, and facilitate monitoring of public policies.
- Create a permanent mechanism for evaluating public satisfaction with digital public services through accessible forms embedded in existing online channels. The results should be analyzed periodically and used to correct problems in the services offered, making municipal management more responsive to citizens' needs.